



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-2027

Course Handout Part II

Date: 01/08/2026

In addition to part -I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : **ME F340**
Course Title : ***Introduction to Sport Engineering***
Instructor-in-Charge : **M. S. Dasgupta**
Practical Instructors : **Mr Bhavesh Verma (SWD), Mr Ajay Shekhawat (SWD)**

1. Course Description:

The course essentially prepares engineering students to utilize their existing technical knowhow for sports applications. The specific topics will include, working with human and human movement patterns, qualitative and quantitative techniques for analysis of human movement. Sports performance measure and analysis. Anthropomorphy, Ergonomics and Task design. Computer application in sports, computerized performance measure. Sports surface and its impact on sports performance, surface testing methods. Sports and training equipment case study, business around sports

2. Scope & Objective:

This is an introductory course to impart knowledge and skill set related to working with human in sports engineering application. Understanding human movement pattern and performance measurement. Understand, assess and analyze effect of equipment, sports surfaces and environment on sports performance. Use of ergonomic concepts, image processing, smart sensors and data analysis in sports engineering. Standard techniques of evaluation and testing of sports equipment and sports surfaces and appreciation of business around sports industry.

3. Learning outcome: Course Learning Outcomes:

By the end of this course, students will be able to:

Understand the fundamentals of human movement and apply engineering principles to analyze sports performance.

Appreciate the role of ethics, human factors, and design considerations in sports technology.

Utilize modern computing tools for motion analysis and performance evaluation in sports.

Gain insights into sports infrastructure, surface design, and their impact on athlete performance.

Explore the business, innovation, and marketing potential within the sports ecosystem from an engineering perspective.

4. Text Books:

(T1) Sport and exercise biomechanics - P. Grimshaw, New York: Taylor & Francis, 2019 2nd Ed,



(T2) Introduction to Sports Biomechanics: Analysing Human Movement Patterns, 2nd edition by Roger Bartlett, Routledge Publishing, 2014

5. **Reference Books:**

(R1) Materials in sports equipment – Mike Jenkins, Woodhead Publishing 2003 UK

(R2) Kinesiology: Scientific basis of human motions, By Katharina F Wel's and Kathryn Luttgens, 9th Edition, Philadelphia

(R3) The Science and Engineering of Sport Surfaces Edited by Sharon Dixon, Paul Fleming, Iain James, Matt Carré

(R4) Database systems: a practical approach to design, implementation and management (5th edition), Connolly, Thomas M. and Carolyn E. Begg, Addison-Wesley 6th Edition, 2015

(R5) *The Routledge Handbook of Ergonomics in Sport and Exercise* Youlian Hong, 2014 Routledge

6. **Course Plan:**

Week no.	Module	Lect. no.	Reference	Learning outcomes
1	Sports and Technology Introduction to Sports engineering and sustainability Objectives and scopes of sports engineering Connecting engineering with sports development.	1-3	T1 Ch:1,2 R4 Sec:1,2	Learning fundamental skills that Engineering students require to understand sports movements. Technology and sports ethics.
2,4	Human Movement Patterns Defining human movements Some fundamental movements and Movement patterns, qualitative and quantitative methods, Gait analysis	4-12	T2 Sec:1,2,3,4	Learning human movement patterns and their analysis
5,6	Ergonomics and Anthropometry Introduction to ergonomics, system design and task analysis Introduction to anthropometry and its application in sports. Anthropometry measures and anthropometric techniques	13-21	R5 Sec:1,2,5 T1 Sec:2,3	Learning basic human factors in engineering design
7	Performance Analysis of Sports Movements What is Performance Analysis of Sport? Quantitative and Qualitative Analysis, Sports Performance Data and Information Computer application in sports Computerized Performance Analysis Systems and AI	22-25	T1 Sec:1,2,4,6	Learning modern computing technology and its use in sports performance analysis
5	Business around Sports Introduction to sports as a product and a service. Pricing and promotion strategies in sports marketing.	26-29	Reading material will be supplied	Learning about marketing opportunities of sporting events and business around sports.

6	Sports Infrastructure and Surfaces Basics of sports Infrastructure. Planning design and management of sports infrastructure. History and development of sports Surfaces, surface Classification and characterization, surface test methods. Sports Surfaces and Performance. Chemistry of Sports Surface.	30-32	R3 Sec:1,2,3 R5 Sec:6	Learning key aspect of sports infrastructure surface development, testing and effect on performance
7	Equipment Case Studies: Materials and design of sports products – balls, tennis rackets, cricket bats, bicycles, running shoes, pole vaults, surfaces and training equipment case studies and measurements	33-35	R1 Sec:8	Visualize situations that engineers need to analyze and appreciate in Sports engineering problem and scope of business around sports

7. Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Mid Semester Test	30	60 Min	-	CB
Hands-on session*	15		Hands-on session hours	OB
Individual project	15		Take-home	OB
Comprehensive Examination	40	120 Min.		CB

* Tentatively, weekly one Hands-on session will be conducted. Attendance and performance in 4 best marks in hands-on sessions will be counted towards evaluation.

8. Criteria for NC: Below 10% of course topper's marks

9. Chamber Consultation Hour: To be announced by individual instructors

10. Notices: Online at Nalanda

11. Make-up Policy: No Make-up is permissible in Hands-on session (evaluated or non-evaluated). Make-up request for Mid Semester / Comprehensive examination must accompany appropriate supporting medical / exigency documents.

[Mani Sankar Dasgupta]
Instructor-in-charge [ME F340]

